

Samadrita Roy

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PROFESSIONAL SUMMARY

Frontend Developer with 4+ years of experience at Tata Consultancy Services delivering **enterprise-scale web applications** across aviation, fintech, and government domains. Specialized in **React.js, TypeScript, scalable component architecture, and API-driven systems**, with proven impact in **performance optimization (~20% improvement)**, **developer efficiency (~70% faster delivery)**, and **accessibility (~90% WCAG audit scores)**. Experienced in building **large-scale production systems supporting 750K+ users**, with strong focus on **NFRs including scalability, security, maintainability, and accessibility**. Currently expanding into **Generative AI and LLM-powered frontend applications**.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript (ES6+), TypeScript, React.js, React Native

State Management & Data: Redux, GraphQL, REST APIs

Architecture & Design: Component Architecture, Scalable UI Systems, Modular Design, NFR Implementation

Performance & Accessibility: Lazy Loading, Render Optimization, WCAG 2.0/2.1, Cross-Browser Compatibility

Tools & Platforms: Git, Storybook, VS Code, Strapi CMS

Testing: Unit Testing

Other: Agile/Scrum, Responsive Design, Pixel-Perfect UI Development

Emerging: Generative AI, Prompt Engineering, LLM Integration

PROFESSIONAL EXPERIENCE

Frontend Developer

Feb 2022 – Present

Tata Consultancy Services

Kolkata, India

- Engineered and delivered **5+ enterprise-grade web applications** across aviation, fintech, and government sectors, supporting **750K+ users in production environments**.
- Improved application performance by **~20%** through implementation of **lazy loading, optimized rendering strategies, and efficient API consumption patterns**.
- Designed and implemented **reusable React component architecture**, accelerating development speed by **~70%** and improving code maintainability at scale.
- Built scalable frontend systems using **TypeScript, Redux, and GraphQL**, ensuring predictable state management and efficient data flow.
- Developed **WCAG-compliant accessible UI systems**, achieving **~90%** audit scores and improving usability for assistive technologies.
- Delivered **pixel-perfect, responsive UI systems** ensuring consistent UX across devices, browsers, and global users.
- Implemented **Non-Functional Requirements (NFRs)** including performance, scalability, security, accessibility, and maintainability across multiple production applications.
- Collaborated in **Agile/Scrum teams**, working closely with backend engineers, designers, and product stakeholders to deliver high-quality releases.

KEY PROJECTS

Department of Social Protection (DSP), Ireland – Pension Platform

Nov 2024 – Present

React, TypeScript, GraphQL, Strapi CMS, React Native

- Contributing to a **government pension platform used by 750K+ users**, enabling enrollment, opt-in/opt-out workflows, and employer/member portal functionality.
- Built **dynamic multilingual UI architecture** using Strapi CMS, creating a custom **BlocksRenderer component** to render API-driven UI content.
- Introduced and implemented **internationalization (i18n)**, enabling scalable localization across regions.
- Worked extensively with **GraphQL APIs** for efficient data fetching and structured communication.
- Ensured **WCAG accessibility compliance** and implemented enterprise-grade NFRs across the platform.
- Currently **developing the complete mobile application single-handedly using React Native**, targeting production release.
- Received recognition from **TCS BANCS and TCS Interactive leadership** for impactful contributions.

Dibber – Education Platform (Norway)

Jan 2024 – Sep 2024

React, Redux, Storybook

- Led frontend development for a **back-office platform used across 600+ schools in 9 countries**, supporting large-scale educational operations.
- Designed and implemented **pixel-perfect, responsive UI systems** ensuring cross-device compatibility.
- Built and maintained a **Storybook-based reusable component library**, improving UI consistency and developer productivity.
- Implemented **Redux-based state management** to handle complex application workflows and data states.
- Applied **WCAG 2.0 accessibility standards**, ensuring inclusive user experience.
- Improved performance using **lazy loading and render optimization techniques**.
- Defined and implemented **Non-Functional Requirements (performance, scalability, security, accessibility, maintainability)**.
- Delivered **unit-tested, production-ready features** with high reliability.
- Received **direct client appreciation** and recognition from **Global Head of TCS Interactive**.

OMSA, Air India Express & Standard Bank – Enterprise Platforms

Feb 2022 – Nov 2024

React, TypeScript, REST APIs

- Delivered frontend solutions across **insurance (OMSA), airline booking (Air India Express), and banking (Standard Bank)** platforms.
- Built **API-driven UI systems**, reusable React components, and **responsive pixel-perfect interfaces**.
- Developed **booking workflows, business logic integrations**, and scalable UI modules.
- Contributed to **cross-domain enterprise systems** (fintech, aviation, government).

PROJECTS

AI Agent (GitHub)

React, Generative AI, LLaMA3, Prompt Engineering

- Developed a **GenAI-powered AI agent** leveraging **LLaMA3 via Ollama**, enabling intelligent task execution through prompt-driven workflows.
- Working on integrating AI workflows with **React-based frontend systems**, enabling experimentation with real-time AI-driven UI interactions.

EDUCATION

Calcutta University

Aug 2018 – Oct 2020

M.Sc in Electronic Science

Kolkata, India

ACHIEVEMENTS

- Recognized by **Global Head of TCS Interactive** for outstanding contributions on Dibber platform.
- Awarded **“Star of the Month” (2x)** for high-impact performance on DSP project.
- Appreciated by **TCS BANCS and TCS Interactive leadership** for enterprise-scale delivery.